

Binance Futures Order Bot

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Introduction:

This project implements an **interactive command-line trading bot** for **Binance USDT-M Futures Testnet** using Python.

The bot allows users to **interactively choose order types, sides, quantities, and prices**, simulating real trading workflows in a safe test environment.

The project emphasizes:

- User interaction
- Input validation
- Modular architecture
- Logging and traceability

Objective:

The objectives of this project are:

- To build an **interactive CLI-based trading system**
- To support **Market, Limit, and Stop-Limit** orders
- To validate user inputs before order placement
- To log all trading actions and errors
- To follow professional software engineering practices

Technology Stack:

Component	Technology
Language	Python
API	Binance Futures USDT-M Testnet
Library	python-binance
Interface	Command Line Interface (CLI)
Logging	Python logging module

Project Architecture:

Trading-Bot/

```
|— src/
|   |— client.py
|   |— market_orders.py
|   |— limit_orders.py
|   |— validation.py
|   |— advanced/
|       |— stop_limit.py
|— run_bot.py
|— README.md
|— report.pdf
|— requirements.txt
```

- **client.py** → initializes Binance client
- **market_orders.py** → market order logic
- **limit_orders.py** → limit order logic
- **stop_limit.py** → advanced stop-limit order
- **run_bot.py** → interactive CLI interface

Interactive Execution Flow:

The bot runs in an interactive loop where the user selects the order type, enters required parameters, and confirms the order before execution.

Sample interaction:

===== BINANCE FUTURES TESTNET BOT =====

1. Market Order
2. Limit Order
3. Stop-Limit Order
4. Exit

Choose option: 2

Side (BUY / SELL): BUY

Quantity: 0.01

Limit price: 25000

Confirm order? (yes/no): yes

☒ Limit order placed

```
Enter symbol (BTCUSDT): BTCUSDT

===== BINANCE FUTURES TESTNET BOT =====
1. Market Order
2. Limit Order
3. Stop-Limit Order
4. Exit
Choose option (1-4): 2
Side (BUY / SELL): BUY
Quantity: 0.01
Limit price: 25000
Confirm order? (yes/no): yes
☒ Limit order placed

===== BINANCE FUTURES TESTNET BOT =====
1. Market Order
2. Limit Order
3. Stop-Limit Order
4. Exit
Choose option (1-4): 4
Exiting bot...
```

Order Types Implemented:

Market Order

- Executes immediately at market price

Limit Order

- Executes at user-defined price

Stop-Limit Order (Advanced)

- Triggered when stop price is reached
- Places limit order after trigger

Validation & Logging:

Input validation ensures that symbol format, quantity, and price values are correct before sending API requests.

All actions, including successful orders and errors, are logged into bot.log with timestamps.

≡ bot.log

```
1 2025-12-30 20:49:45,947 - INFO - {}
2 2025-12-30 21:46:59,651 - INFO - Market order: {}
3 2025-12-30 22:34:56,501 - INFO - Limit order placed: {}
```

Security Measures:

- API keys stored in config.py
- config.py excluded using .gitignore
- Uses Binance **Testnet only**
- No real funds used

Conclusion:

This project demonstrates the development of a real-world, interactive trading bot using the Binance Futures Testnet. It follows modular design principles, supports advanced orders, and includes validation and logging for safe and reliable operation.